

Communication with I/O Devices

Overview

The CPU communicates with external devices like keyboards, displays, printers, disk drives, and network cards through an I/O subsystem. Efficient communication is crucial for system performance.

I/O Communication Techniques

Programmed I/O (Polling)

- The CPU manually checks (polls) the device until it's ready.
- **Steps:**
 1. CPU sends a command to the device.
 2. CPU enters a loop and repeatedly checks device status.
 3. When ready, CPU transfers data.
- **Pros:** Simple and easy to implement.
- **Cons:** CPU remains busy and inefficient, especially with slow devices.

Interrupt-Driven I/O

- Devices interrupt the CPU when they are ready.
- **Steps:**
 1. CPU sends a request and continues executing other tasks.
 2. Device sends an interrupt signal when ready.
 3. CPU halts its task and runs the ISR (Interrupt Service Routine).
 4. After servicing, the CPU resumes its previous task.
- **Pros:** Efficient CPU usage.
- **Cons:** Requires interrupt handling logic and ISR.

Direct Memory Access (DMA)

- Used for high-speed devices like disks and network cards.
- DMA controller transfers data between memory and the I/O device.
- CPU only initializes the process and is free during transfer.
- **Pros:** Fast and efficient.
- **Cons:** Requires additional DMA hardware.

Mode	Controller	CPU Free?	Speed	Best Use Case
Programmed I/O	CPU	✗	Slow	Simple devices
Interrupt-Driven I/O	I/O Device	✓	Medium	Keyboard, Mouse
DMA	DMA Controller	✓✓	Fast	Disk, Audio, Network

Table 1: Comparison of I/O Data Transfer Modes

Data Transfer Modes Comparison

I/O Addressing Techniques

Memory-Mapped I/O

- I/O devices share the same address space as RAM.
- Devices can be accessed using regular load/store instructions (e.g., `MOV`, `LOAD`).
- **Pros:** Simpler instruction set, fewer I/O-specific instructions.
- **Cons:** Reduces available memory address space.

I/O-Mapped (Isolated) I/O

- I/O devices have a separate address space.
- CPU uses special instructions (e.g., `IN`, `OUT`) to access devices.
- **Pros:** Full memory address space is preserved.
- **Cons:** Requires additional I/O instructions.

Visual Summary

- **Communication Modes:**
 - **Programmed I/O** – CPU polls the device.
 - **Interrupt-Driven I/O** – Device signals CPU via interrupt.
 - **DMA** – External controller handles transfer independently.
- **Addressing Techniques:**
 - **Memory-Mapped I/O** – Same space as RAM.
 - **I/O-Mapped I/O** – Separate address space.

Additional Notes

- **ISR (Interrupt Service Routine):** Small software routine to handle device interrupt.
- **CPU Efficiency:** Interrupts and DMA are designed to free CPU time for computation.
- **Real-Life Analogy:**
 - **Polling:** Repeatedly checking your mailbox.
 - **Interrupt:** Postman rings your doorbell when mail arrives.
 - **DMA:** Hiring someone to pick up and deliver your mail directly.